Agreement-Based Segment Discovery with Joint Segment-Ratio Reinforcement Learning

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Abstract

Reinforcement learning (RL) increasingly trains one language model across heterogeneous reasoning domains, whose responses can differ substantially in temporal structure. Token-level objectives can assign inconsistent importance weights and clipping decisions within one reasoning state, whereas sequence-level objectives such as GSPO treat the entire response as one unit and cannot adapt to state changes within it. We propose AGREEMENT-RL, a domain-agnostic two-stage framework that discovers contiguous segments from the agreement of advantage-weighted token updates and then optimizes each segment with its exact joint likelihood ratio. A self-excluded resultant score retains pairwise token interactions, and a dynamic program finds the globally optimal partition. Across seven benchmarks and two Qwen3.5 model scales, AGREEMENT-RL improves pass@1 over the matching token-level baseline in all 28 model–algorithm–task comparisons. Averaged over GRPO- and PPO-style instantiations, the gains on mathematical and in-domain multi-hop reasoning are 5.9 and 2.2 points for the 9B model, and 5.7 and 2.0 points for the 4B model; the average gain on the out-of-domain Bamboogle benchmark is 2.8 points. Performance is also stable across the tested projection dimensions and candidate-length caps.

1 Introduction

Reinforcement learning has become a standard way to improve the reasoning and instruction-following abilities of large language models (LLMs) beyond supervised fine-tuning (Ouyang et al. 2022; Shao et al. 2024; Guo et al. 2025). A growing objective is to train one policy across multiple domains rather than maintain a separate model for every task family. Multi-domain RL can share reasoning skills and amortize training cost, but it also combines reward landscapes with different difficulty, response structure, and preferred parameter updates.

The central optimization challenge is gradient conflict. An update that benefits mathematical deduction can be neutral or harmful to search-intensive multi-hop reasoning, and even related domains may disagree at particular training stages. Multi-task optimization methods diagnose or modify such conflicts using gradient magnitude and direction (Chen et al. 2018; Sener and Koltun 2018; Yu et al. 2020; Liu et al. 2021). These methods generally treat each task update as a single vector. Language-model RL adds a sequential question: *where do token updates form a coherent reasoning state?* A response may move through planning, evidence gathering, calculation, and answer synthesis, so its locally appropriate update granularity can change within one trajectory.

Fixed temporal abstraction does not resolve this issue: distributing an update over tokens, fixed spans, or a complete response imposes one granularity despite heterogeneous response structures. Section 2 reviews this distinction and motivates an adaptive notion of update state.

We propose AGREEMENT-RL, which discovers contiguous states by asking where advantage-weighted token updates reinforce one another. Candidate spans are scored by their pairwise token-update agreement, and dynamic programming selects the globally optimal partition. Each discovered segment is then optimized as a macro-action with its exact joint likelihood ratio, so all tokens in the state share one importance weight and clipping decision. The method requires only a standard response-level advantage and no domain-conditioned verifier outputs. Algorithm 1 gives the procedure.

Our evaluation uses Qwen3.5 models at 9B and 4B scales (Qwen Team 2026). We compare AGREEMENT-RL with matching token-level GRPO and PPO baselines, GSPO-based variants as sequence-level comparisons, and Proxy-PCGrad as a conflict-aware multi-task baseline. AGREEMENT-RL improves pass@1 on every benchmark under both RL algorithms, yielding 28 gains across 28 model–algorithm–task comparisons. Relative to the token-level baselines and averaged across the two algorithms, it improves mathematical reasoning by 5.9 points at 9B and 5.7 points at 4B, while improving in-domain multi-hop reasoning by 2.2 and 2.0 points. On Bamboogle, the four model–algorithm gains range from 0.8 to 4.8 points and average 2.8 points.

Our contributions are as follows:

- We introduce AGREEMENT-RL, which discovers contiguous response states through token-update agreement and uses each discovered segment as a joint RL update unit, replacing tokenizer-defined or fixed-chunk granularity.
- We derive the agreement score's pairwise interpretation and an efficient prefix-sum dynamic program that returns the globally optimal partition.

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- We evaluate AGREEMENT-RL across seven benchmarks, two model scales, and GRPO- and PPO-style implementations, demonstrating consistent in-domain and OOD improvements over token-level, sequence-level, and conflict-aware baselines, with sensitivity and runtime analyses of segment discovery.

2 Background

2.1 Temporal Abstraction as an MDP Design Choice

Language-model RL can represent generation at several temporal resolutions. A token-level formulation treats each next-token prediction as an action. Trajectory-level formulations such as GSPO treat the complete response as one composite action (Zheng et al. 2025b), while fixed-length windows define intermediate macro-actions. These abstractions determine how likelihood correction, trust-region control, and credit are shared across time, but they do not determine how signals from different domains should be reconciled.

The distinction between token- and trajectory-level optimization is also less fundamental than it first appears. Zheng et al. show that a token-level importance-weighted surrogate can approximate the expected sequence-level reward objective to first order when the rollout and target policies remain close and the training and inference engines agree (Zheng et al. 2025a). Token-level optimization can therefore retain the intended trajectory objective without explicitly using a trajectory ratio. Conversely, directly optimizing sequence likelihood is difficult because its numerical range and variance grow across long responses. Fixed windows trade between these behaviors, but their boundaries remain an external modeling choice.

More importantly, multi-domain training combines responses whose useful update states need not share one temporal scale. Token-level optimization can fragment a coherent reasoning operation, while response-level aggregation can merge planning, retrieval, calculation, and synthesis even when their token updates point in different directions. This motivates discovering spans from within-response update coherence rather than selecting token-, window-, or trajectory-level granularity in advance.

2.2 Previous Methods for Multi-Domain Conflict Resolution

Existing multi-task optimization methods address heterogeneous objectives from several complementary perspectives. Representation-sharing approaches such as Distral regularize task policies toward a distilled common policy (Teh et al. 2017), while PopArt and GradNorm correct reward, loss, or gradient-scale imbalance (Hessel et al. 2019; Chen et al. 2018). Multi-objective approaches seek a compromise direction, including MGDA's common descent solution and CA-Grad's trade-off between average progress and worst-case conflict (Sener and Koltun 2018; Liu et al. 2021). These methods improve transfer or balance, but a shared policy may still receive directionally incompatible updates after their scales have been normalized.

PCGrad is among the most widely adopted representatives of gradient surgery (Yu et al. 2020). Its appeal comes from a simple, optimizer-agnostic rule: when two task gradients conflict, remove the component of each gradient that opposes the other before aggregation. It requires neither an auxiliary task-specific network nor a constrained solver, and has shown broad effectiveness across multi-task learning and reinforcement-learning settings. These properties make PCGrad a natural conflict-aware baseline for multi-domain language-model RL.

PCGrad nevertheless treats each task as one aggregate direction, averaging away conflict that varies across planning, retrieval, reasoning, and synthesis before correction. Exact task gradients are also expensive to materialize in distributed LLM training, while approximate surgery can leave a weak residual when task directions are nearly opposed. Thus PCGrad modifies the global update without identifying the temporal spans over which conflict occurs.

3 Preliminaries

Multi-domain rollouts. Let B be the rollout batch size. Each prompt x_i is drawn from a mixture of training domains, and the old policy π_{θ_0} samples a response $y_i = (y_{i,1}, \ldots, y_{i,T_i})$ with T_i valid response tokens. The current policy is π_θ. We index responses by i and token positions by t. Padding, prompt, and truncated positions are excluded; domain labels are not required by the method.

Token likelihood ratios. The current-to-old token log-ratio is

$$\delta_{i,t}(\theta) = \log \pi_\theta(y_{i,t} \mid x_i, y_{i,<t}) - \log \pi_{\theta_0}(y_{i,t} \mid x_i, y_{i,<t}). \tag{1}$$

Advantage-weighted token updates. Each rollout has one response-level advantage $\widehat{A}_i$, which is broadcast to every valid response token. The ideal local update direction is

$$g_{i,t} = \widehat{A}_i \nabla_\theta \log \pi_{\theta_0}(y_{i,t} \mid x_i, y_{i,<t}). \tag{2}$$

Valid segmentations. Each response is partitioned into an ordered sequence of disjoint, contiguous spans that covers every valid response token exactly once. We denote this partition by $\mathcal{S}_i$, its k-th span by $S_{i,k}$, the number of spans by K_i, and the set of feasible partitions by $\mathfrak{S}_i$. Segmentation is discovered from old-policy rollouts and treated as fixed during optimization on that batch.

4 Method

4.1 Agreement Score for a Candidate Segment

The purpose of segmentation is to identify a contiguous span whose token updates behave as one coherent state. For a candidate segment $S = [l, r]$, let its accumulated ideal update be

$$G_i(S) = \sum_{t=l}^{r} g_{i,t}. \tag{3}$$

We remove the diagonal self-strength from the squared resultant and define

$$\mathcal{A}_i(S) = \|G_i(S)\|_2^2 - \sum_{t=l}^{r} \|g_{i,t}\|_2^2 = 2 \sum_{l \le t < u \le r} g_{i,t}^\top g_{i,u}. \quad (4)$$

The second line follows by expanding the squared norm. Aligned token updates contribute positively, orthogonal updates contribute zero, and opposing updates reduce the score. Subtracting the diagonal prevents isolated update magnitude from being mistaken for agreement. A boundary is useful when the agreement retained on its two sides exceeds that of their union; the final decision is nevertheless global rather than based on a local threshold. Because $\widehat{A}_i$ is constant within a response, it scales every candidate and partition score by $\widehat{A}_i^2$. Hence nonzero reward scale and sign do not change the selected boundaries, while a zero-advantage rollout contributes no actor update. Figure 1 illustrates the local merge–split criterion induced by Equation (4).

4.2 Globally Optimal Segment Discovery

Independent boundary decisions can be mutually inconsistent. We instead score a partition additively:

$$Q(\mathcal{S}_i) = \sum_{k=1}^{K_i} \mathcal{A}_i(S_{i,k}), \qquad \mathcal{S}_i^\star = \arg\max_{\mathcal{S}_i \in \mathfrak{S}_i} Q(\mathcal{S}_i), \quad (5)$$

where $\mathfrak{S}_i$ is the set of feasible partitions.

Define $F_i(r)$ as the best score for prefix $[1, r]$, with $F_i(0) = 0$ and unreachable states equal to $-\infty$. Optimal substructure yields

$$F_i(r) = \max_{1 \le l \le r} \{F_i(l-1) + \mathcal{A}_i([l, r])\}. \quad (6)$$

A formal induction proving global optimality appears in Appendix C. Practical boundary recovery and candidate caching are described in Section 4.5.

4.3 Joint Segment-Ratio Policy Optimization

Once discovery fixes the boundaries, each segment is treated as one macro-action. Autoregressive factorization requires its log-ratio to add the constituent token log-ratios:

$$\Delta_{i,k}(\theta) = \sum_{t \in S_{i,k}} \delta_{i,t}(\theta), \qquad \rho_{i,k}(\theta) = \exp(\Delta_{i,k}(\theta)). \quad (7)$$

Thus $\rho_{i,k}$ is the exact probability ratio of the sampled segment conditional on the prompt and its preceding prefix. Because the response advantage is broadcast, every discovered segment uses $\widehat{A}_i$. For $0 < \epsilon < 1$, the clipped surrogate is

$$J(\theta) = \frac{1}{B} \sum_{i=1}^{B} \frac{1}{T_i} \sum_{k=1}^{K_i} \min\big(\rho_{i,k}(\theta)\widehat{A}_i, \operatorname{clip}(\rho_{i,k}(\theta), 1-\epsilon, 1+\epsilon)\widehat{A}_i\big). \quad (8)$$

The training loss is $-J(\theta)$. Normalizing by the original token count T_i, rather than the number of segments K_i, preserves the token-level baseline's initial policy gradient. After the policy moves away from θ_0, tokens in a segment share the joint ratio and clipping decision. This isolates aggregation granularity while retaining the baseline normalization; Appendix D gives the derivation.

4.4 Practical Gradient Approximation

Since the true gradient $g_{i,t}$ in Equation (2) is difficult to obtain in practice, we instead use a proxy gradient computed at the logits. Let $\mathcal{V}$ be the model vocabulary, $z_{i,t} \in \mathbb{R}^{|\mathcal{V}|}$ the old policy's logits, $\boldsymbol{\pi}_{i,t} = \operatorname{softmax}(z_{i,t})$ their probability vector, and $e_{y_{i,t}} \in \mathbb{R}^{|\mathcal{V}|}$ the one-hot vector of the sampled token. The advantage-weighted logit proxy is

$$\xi_{i,t} = \widehat{A}_i(e_{y_{i,t}} - \boldsymbol{\pi}_{i,t}). \quad (9)$$

This follows from the closed-form log-softmax gradient and requires only forward-pass probabilities (Li et al. 2026). Let p be the target projection dimension. Because $\xi_{i,t}$ has vocabulary dimensionality, after fixing each rollout batch we draw one Gaussian matrix shared by all of its tokens and candidate spans,

$$R \in \mathbb{R}^{p \times |\mathcal{V}|}, \qquad R_{ab} \sim \mathcal{N}(0, 1/p), \quad (10)$$

and use the projected practical feature

$$\widetilde{g}_{i,t} = R\xi_{i,t} \in \mathbb{R}^p. \quad (11)$$

In implementation, $\widetilde{g}_{i,t}$ replaces $g_{i,t}$ throughout the agreement score and dynamic program.

Geometry preservation. OPT establishes that the logit proxy norm is proportional to the corresponding last-layer gradient norm (Li et al. 2026); our score additionally requires cross-token inner products. For a candidate span, define its accumulated unprojected and projected proxies as

$$U_i(S) = \sum_{t \in S} \xi_{i,t}, \qquad \widetilde{G}_i(S) = RU_i(S). \quad (12)$$

Let $J_{i,S}$ be a representative logit Jacobian for span S. Given $J_{i,S}$, define the token residuals by the exact decomposition

$$g_{i,t} = J_{i,S}^\top \xi_{i,t} + r_{i,t|S}, \qquad t \in S. \quad (13)$$

We then make two modeling assumptions. First, the representative Jacobian is locally conformal:

$$J_{i,S} J_{i,S}^\top = c_{i,S} I, \qquad c_{i,S} > 0. \quad (14)$$

Second, within-span Jacobian variation produces bounded relative token residuals:

$$\|r_{i,t|S}\|_2 \le \kappa \sqrt{c_{i,S}} \|\xi_{i,t}\|_2, \qquad t \in S, \qquad \kappa \ge 0. \quad (15)$$

Here, $c_{i,S}$ is the local scale and κ bounds relative Jacobian variation. Conditioned on the rollout batch, assume its M nonzero token proxies are fixed independently of R. Let $\epsilon_{\mathrm{JL}} \in (0, 1)$ be the desired projection distortion and $\eta \in (0, 1)$ the failure probability. For a universal constant C, choose

$$p \ge C \epsilon_{\mathrm{JL}}^{-2} \log(M/\eta). \quad (16)$$

$$\mathcal{A}_i(L \cup R) - \mathcal{A}_i(L) - \mathcal{A}_i(R) = 2\, G_i(L)^\top G_i(R)$$

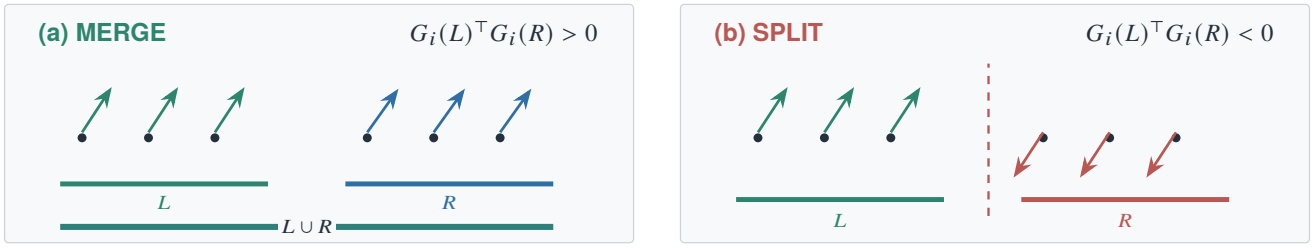


Figure 1: When adjacent candidate spans L and R are combined, the score changes by $2G_i(L)^\top G_i(R)$. Aligned accumulated updates make this cross-boundary term positive and favor merging; opposing updates make it negative and favor a finer split. A zero inner product gives a local tie. The dynamic program applies this comparison globally across all boundaries, subject to the candidate-length constraints.

Proposition 1. Define the projected score by $\widetilde{\mathcal{A}}_i(S) = \|\widetilde{G}_i(S)\|_2^2 - \sum_{t\in S} \|\widetilde{g}_{i,t}\|_2^2$. Under the assumptions in Equations (14) and (15) and the projection condition in Equation (16), with probability at least $1-\eta$, every candidate span satisfies

$$\left|\widetilde{\mathcal{A}}_i(S) - c_{i,S}^{-1}\mathcal{A}_i(S)\right| \leq 2(\epsilon_{\mathrm{JL}} + 2\kappa + \kappa^2) \sum_{\substack{t,u\in S \\ t<u}} \|\xi_{i,t}\|_2 \|\xi_{i,u}\|_2. \tag{17}$$

Thus the proxy preserves the ideal pairwise agreement score whenever the right-hand side is small relative to its scale. Appendix B gives the derivation.

Practical validity. The assumptions are motivated by the local structure used in LLM training. Transformer positions share policy parameters, normalization encourages comparable representation scales, and the candidate-length cap $L_{\max}$ limits each span's positional extent relative to an unrestricted response-level update. We use one Gaussian projection of fixed dimension p, shared across all tokens and candidate spans in each rollout batch, so every agreement comparison is performed in the same projected space. For any fixed proxy vectors u, v, it also preserves their inner product in expectation: $\mathbb{E}_R[\langle Ru, Rv\rangle] = \langle u, v\rangle$. Equation (16) gives sufficient asymptotic scaling, but its universal constant and the realized number of candidate vectors are not estimated during training; we therefore use a fixed projection dimension across all experiments to control projection cost consistently. A direct model-specific proof throughout training is impractical because evolving parameters and optimizer states continually change the Jacobian geometry; materializing exact per-token gradients would also require repeated backward passes or custom hooks that drastically reduce actor-update speed. The consistent downstream improvements indicate that the resulting approximation provides useful agreement signals in practice.

4.5 Training Procedure and Complexity

Algorithm 1 summarizes discovery and optimization with the practical features. Prefix sums cache accumulated updates and squared norms, allowing every candidate score to be formed in constant time after projection. The dynamic program stores the best predecessor for each response prefix; following these backpointers from the final token recovers the selected segments. This recovery is an implementation step and does not change the partition objective.

Minimum and maximum span lengths may be supplied as optional implementation controls. They restrict candidate enumeration and regulate the variance of joint ratios, but are deliberately omitted from the formal recurrence. Without a maximum span length, candidate enumeration is quadratic in response length. With a maximum candidate length, the candidate-scoring time, prefix-cache storage, and dense-projection work per response are, respectively,

$$\begin{aligned} \mathcal{C}_{\mathrm{score}}(i) &= O(T_i L_{\max} p), \\ \mathcal{M}_{\mathrm{cache}}(i) &= O(T_i p), \\ \mathcal{C}_{\mathrm{proj}}(i) &= O(T_i p |\mathcal{V}|). \end{aligned} \tag{18}$$

The projection cost is incurred once per rollout. Ratios and clipping are evaluated in log space to avoid numerical overflow.

In implementation, we use batched parallel computation and GPU acceleration for projection, candidate scoring, and dynamic-program updates.

5 Experiments

5.1 Experimental Setup

Training data. We train jointly on two balanced domains with 9,000 instances in total. The multi-hop domain combines 2WikiMultiHopQA (Ho et al. 2020), MuSiQue (Trivedi et al. 2022), and HotpotQA (Yang et al. 2018). We sample 1,500 training examples uniformly at random from each source, yielding 4,500 multi-hop instances. During rollouts, the policy accesses the wikipedia Python package as its search tool (Goldsmith 2014). The mathematical domain contains 4,500 randomly sampled examples from BytedTsinghua-SIA/DAPO-Math-17k (Yu et al. 2025). Balanced domain sampling prevents the larger source from determining the update frequency.

Method	Metric	MATH L5	AMC	Numina	2Wiki *@1 / @2 / @4*	MuSiQue	HotpotQA	Bamboogle
GRPO	pass@k	79.0 / 85.6 / 89.6	61.4 / 68.7 / 71.1	38.0 / 45.4 / 50.4	29.0 / 36.4 / 42.4	9.2 / 11.6 / 14.2	24.8 / **30.0** / 31.0	43.2 / 49.6 / 53.6
	avg@k	79.0 / 79.1 / 78.9	61.4 / 61.5 / 61.3	38.0 / 38.1 / 37.9	29.0 / 28.9 / 29.1	9.2 / **10.3** / **10.2**	24.8 / 24.7 / 24.3	43.2 / **43.3** / **43.4**
Proxy-PCGrad	pass@k	76.0 / 83.8 / 89.0	59.0 / 67.5 / 72.3	34.6 / 43.4 / 49.8	26.2 / 35.0 / 42.6	7.0 / 10.6 / **14.6**	21.8 / 28.4 / 30.8	40.8 / 48.8 / 54.4
	avg@k	76.0 / 75.9 / 76.1	59.0 / 59.1 / 58.9	34.6 / 34.7 / 34.5	26.2 / 26.3 / 26.1	7.0 / 6.9 / 7.1	21.8 / 22.5 / 22.4	40.8 / 40.7 / 41.7
GSPO	pass@k	80.4 / 87.4 / **91.4**	62.7 / 69.9 / 73.5	42.4 / 49.8 / 56.0	29.4 / 36.8 / 43.0	7.8 / 9.6 / 11.4	23.0 / 25.8 / 29.0	42.4 / 48.0 / 51.2
	avg@k	80.4 / 80.5 / 80.3	62.7 / 62.8 / 62.6	42.4 / 42.3 / 42.5	29.4 / **29.9** / **29.8**	7.8 / 7.7 / 7.0	23.0 / 22.4 / 22.3	42.4 / 43.2 / 41.4
AGREEMENT-RL (GRPO)	pass@k	**83.8** / **87.8** / 91.0	**67.5** / **73.5** / **77.1**	**43.6** / **51.0** / **56.2**	**30.2** / **40.6** / **46.8**	**11.2** / **13.0** / 14.0	**25.4** / 29.8 / **32.8**	**44.0** / **52.0** / **56.8**
	avg@k	**83.8** / **83.7** / **83.9**	**67.5** / **67.6** / **67.4**	**43.6** / **43.7** / **43.5**	**30.2** / 29.8 / 29.7	**11.2** / 10.2 / 10.1	**25.4** / **25.3** / **25.6**	**44.0** / 42.9 / 43.1

Table 1: Per-task 9B GRPO-family validation results from the final checkpoint after training (%). Tasks are columns and methods are rows; each task cell lists @1 / @2 / @4. Source `mean@`k is shown as avg@k. Bold marks the best value for each task, metric, and k.

Method	Metric	MATH L5	AMC	Numina	2Wiki *@1 / @2 / @4*	MuSiQue	HotpotQA	Bamboogle
GRPO	pass@k	70.4 / 79.2 / 84.2	48.2 / 59.0 / 65.1	35.8 / 45.4 / 52.4	26.8 / 30.8 / 33.6	6.0 / 7.0 / 10.0	21.0 / 21.0 / **24.0**	33.6 / **39.2** / 42.4
	avg@k	70.4 / 71.8 / 70.3	48.2 / 48.3 / 48.1	35.8 / 35.7 / 35.9	26.8 / 26.7 / 27.1	6.0 / 5.9 / 5.8	21.0 / **20.9** / 21.1	33.6 / 33.0 / 33.4
Proxy-PCGrad	pass@k	67.4 / 77.4 / 83.6	45.8 / 56.6 / 66.3	32.4 / 43.4 / 51.8	24.0 / 29.4 / 33.8	3.8 / 6.0 / 10.4	17.8 / 19.4 / 23.8	31.2 / 38.4 / 43.2
	avg@k	67.4 / 67.5 / 67.3	45.8 / 45.7 / 45.9	32.4 / 32.5 / 32.3	24.0 / 24.1 / 23.9	3.8 / 3.7 / 5.5	17.8 / 17.7 / 18.2	31.2 / 31.1 / 32.5
GSPO	pass@k	72.0 / 80.2 / **87.2**	49.4 / **60.2** / 67.5	40.8 / 49.2 / 58.6	26.6 / 32.2 / 34.8	5.8 / 6.8 / 8.4	18.6 / 19.2 / 23.0	32.8 / 36.8 / 41.6
	avg@k	72.0 / **72.1** / 71.9	49.4 / 49.5 / 49.3	40.8 / 40.5 / **41.4**	26.6 / 27.1 / **28.6**	5.8 / 5.7 / **5.9**	18.6 / 19.2 / 18.1	32.8 / 33.1 / 31.6
AGREEMENT-RL (GRPO)	pass@k	**73.0** / **82.8** / 86.2	**54.2** / 57.8 / **71.1**	**41.8** / **51.0** / **58.8**	**29.2** / **35.4** / **37.4**	**7.0** / **9.6** / **11.6**	**21.2** / **22.0** / 22.0	**35.2** / 37.6 / **45.6**
	avg@k	**73.0** / 71.7 / **72.9**	**54.2** / **54.1** / **54.3**	**41.8** / **42.1** / 41.1	**29.2** / **28.7** / 27.0	**7.0** / **6.9** / 5.4	**21.2** / 19.1 / **21.3**	**35.2** / **35.3** / **35.1**

Table 2: Per-task 4B GRPO-family validation results from the final checkpoint after training (%). Tasks are columns and methods are rows; each task cell lists @1 / @2 / @4. Source `mean@`k is shown as avg@k. Bold marks the best value for each task, metric, and k.

Method	Metric	MATH L5	AMC	Numina	2Wiki *@1 / @2 / @4*	MuSiQue	HotpotQA	Bamboogle
PPO	pass@k	76.6 / 84.6 / 90.4	57.8 / 66.3 / 72.3	39.4 / 45.6 / 52.0	28.0 / 38.0 / 42.8	10.0 / 13.0 / 16.0	22.8 / **29.6** / 33.0	40.0 / 48.0 / 54.4
	avg@k	76.6 / 76.7 / 76.5	57.8 / 57.7 / 57.9	39.4 / 39.5 / 39.3	28.0 / 29.6 / 27.9	10.0 / 10.1 / **9.9**	22.8 / 22.7 / 23.4	40.0 / 39.9 / **41.3**
Proxy-PCGrad	pass@k	73.6 / 82.8 / 89.8	55.4 / 65.1 / 73.5	36.0 / 43.6 / 51.6	25.0 / 36.6 / 43.0	7.8 / 12.0 / **16.4**	19.8 / 27.8 / 32.8	37.6 / 47.2 / 55.2
	avg@k	73.6 / 73.5 / 73.7	55.4 / 55.5 / 55.3	36.0 / 36.1 / 35.9	25.0 / 25.1 / 24.9	7.8 / 7.7 / 9.6	19.8 / 19.9 / 19.7	37.6 / 37.5 / 41.1
PPO-GSPO	pass@k	79.2 / **87.8** / 93.2	60.2 / 71.1 / 74.7	43.0 / 51.2 / **56.2**	29.0 / 37.8 / 44.8	8.8 / 11.0 / 13.4	21.8 / 27.4 / 31.4	40.8 / 48.8 / 53.6
	avg@k	79.2 / 79.3 / 79.1	60.2 / 60.5 / **62.8**	43.0 / **43.4** / 42.9	29.0 / **30.5** / 29.6	8.8 / 8.7 / 9.5	21.8 / 21.7 / 21.9	40.8 / **42.9** / 41.2
AGREEMENT-RL (PPO)	pass@k	**83.8** / 86.4 / **93.4**	**65.1** / **75.9** / **77.1**	**43.6** / **52.4** / 53.4	**31.2** / **41.0** / **47.8**	**11.4** / **15.4** / 16.2	**27.4** / 28.2 / **35.0**	**44.8** / **54.4** / **57.6**
	avg@k	**83.8** / **83.7** / **83.9**	**65.1** / **64.8** / 62.4	**43.6** / 43.2 / **43.7**	**31.2** / 29.5 / **30.6**	**11.4** / **11.3** / 9.4	**27.4** / **27.3** / **26.8**	**44.8** / 42.5 / 41.0

Table 3: Per-task 9B PPO-family validation results from the final checkpoint after training (%). Tasks are columns and methods are rows; each task cell lists @1 / @2 / @4. Source `mean@`k is shown as avg@k. Bold marks the best value for each task, metric, and k.

Method	Metric	MATH L5	AMC	Numina	2Wiki *@1 / @2 / @4*	MuSiQue	HotpotQA	Bamboogle
PPO	pass@k	68.6 / 78.6 / 85.4	45.8 / 57.8 / 66.3	36.8 / 46.8 / 52.8	26.0 / 31.6 / 35.2	6.4 / 8.0 / 11.6	19.6 / 21.6 / 25.8	31.2 / 38.4 / 44.0
	avg@k	68.6 / 68.5 / 71.7	45.8 / 49.4 / 45.7	36.8 / 36.7 / 36.9	26.0 / 26.1 / 25.9	6.4 / **6.9** / 6.2	19.6 / 19.7 / **19.5**	31.2 / 32.8 / 31.1
Proxy-PCGrad	pass@k	65.6 / 76.8 / 84.8	43.4 / 56.6 / 67.5	33.4 / 44.8 / 52.2	23.2 / 30.0 / 35.4	4.2 / 7.0 / 12.0	16.4 / 19.8 / 25.6	28.8 / 37.6 / 44.8
	avg@k	65.6 / 65.7 / 65.5	43.4 / 43.3 / 43.5	33.4 / 33.5 / 33.3	23.2 / 23.1 / 23.3	4.2 / 4.1 / 5.8	16.4 / 16.3 / 19.0	28.8 / 28.9 / 28.7
PPO-GSPO	pass@k	71.0 / 81.2 / 89.0	47.0 / **61.4** / 69.9	40.4 / 51.8 / **58.2**	26.2 / 32.2 / **37.4**	6.2 / 7.6 / 10.0	17.8 / 20.4 / 25.2	30.4 / 39.2 / 43.2
	avg@k	71.0 / 71.7 / **71.8**	47.0 / **50.2** / 48.8	40.4 / 40.3 / 40.1	26.2 / 27.6 / 28.2	6.2 / 6.8 / **6.3**	17.8 / 17.7 / 19.1	30.4 / **32.9** / 32.2
AGREEMENT-RL (PPO)	pass@k	**74.8** / **84.2** / **89.6**	**53.0** / 59.0 / **73.5**	**42.8** / **54.4** / 55.8	**31.2** / **35.4** / 36.0	**7.4** / **10.4** / **13.8**	**21.8** / **23.2** / **26.8**	**35.2** / **44.0** / **48.0**
	avg@k	**74.8** / **74.1** / 71.6	**53.0** / 49.3 / **51.1**	**42.8** / **42.7** / **43.1**	**31.2** / **29.8** / **29.2**	**7.4** / 6.7 / 5.7	**21.8** / **21.7** / 18.9	**35.2** / 32.5 / **33.4**

Table 4: Per-task 4B PPO-family validation results from the final checkpoint after training (%). Tasks are columns and methods are rows; each task cell lists @1 / @2 / @4. Source `mean@`k is shown as avg@k. Bold marks the best value for each task, metric, and k.

Evaluation data. Mathematical reasoning is evaluated on DigitalLearningGmbH/MATH-lighteval Level 5, AI-MO/aimo-validation-amc, and AI-MO/NuminaMath-CoT (Hendrycks et al. 2021; AI-MO 2024a,b). Multi-hop rea-

Algorithm 1 Agreement Discovery and Segment-Ratio Update

Require: Responses $\{y_i\}$, probabilities $\boldsymbol{\pi}_{i,t}$, advantages $\widehat{A}_i$, matrix R, length bounds

for each response i **do**
  Form $\xi_{i,t}$ and $\widetilde{g}_{i,t} \leftarrow R\xi_{i,t}$
  Build prefix sums of $\widetilde{g}_{i,t}$ and $\|\widetilde{g}_{i,t}\|_2^2$
  $F_i(0) \leftarrow 0$; $F_i(r > 0) \leftarrow -\infty$
  for $r = 1, \ldots, T_i$ **do**
    for each feasible start l **do**
      Accumulate $\widetilde{G}_i([l, r])$ by prefix subtraction
      Score candidate $[l, r]$ using Equation (4)
      Update $F_i(r)$ and its backpointer using Equation (6)
    end for
  end for
  Backtrack to recover $\mathcal{S}_i^\star$
  Sum token log-ratios within each discovered segment
end for
Minimize the loss in Equation (8)

soning is evaluated on held-out examples from 2WikiMultiHopQA, MuSiQue, and HotpotQA, plus Bamboogle as an out-of-distribution search benchmark (Press et al. 2023). Unless otherwise stated, every evaluation set contains 500 examples. Bamboogle contains 125 examples and aimo-validation-amc contains 83.

Baselines. We instantiate vanilla PPO (Schulman et al. 2017) and GRPO (Shao et al. 2024) on Qwen3.5-9B and Qwen3.5-4B models (Qwen Team 2026). For each algorithm and model scale, AGREEMENT-RL uses the same advantages as its matched token-level baseline. We additionally compare against GSPO and PPO-GSPO as sequence-level controls and Proxy-PCGrad as a conflict-aware multi-task baseline using the same practical gradient representation. GSPO uses GRPO-computed advantages with one sequence-level likelihood ratio instead of token-level ratios. PPO-GSPO applies the same sequence-level aggregation while retaining the response advantages computed by PPO.

Implementation details. The lower and upper clipping margins are both set to 0.2. For GRPO, we sample $n = 8$ responses per prompt. For PPO, we use $n = 1$ and set its return parameters to $\lambda = \gamma = 1$. The KL-penalty coefficient is 0.001, the rollout batch size is 32, and the learning rate is 3×10^{-6}. We fix the projection dimension to $p = 64$ and the maximum candidate length to $L_{\max} = 512$ in the main experiments. We train all models for 3 epochs to ensure full convergence. All methods use one response-level advantage per rollout; AGREEMENT-RL broadcasts it to valid tokens for discovery and reuses it for every discovered segment. Tables 1–4 report the validation results of the final checkpoint after training. For search tasks, each interaction turn comprises one model reasoning span followed by one tool call. Following the practical approximation in Section 4.4, we construct advantage-weighted token proxies from rollout probabilities, multiply them by a fresh batch-shared Gaussian random matrix, and perform prefix-sum candidate scoring and dynamic programming in GPU batches. Experiments are conducted on clusters of NVIDIA H200 GPUs with 144 GB memory. The implementation is built with PyTorch (Paszke et al. 2019) and uses veRL for distributed reinforcement-learning training (Sheng et al. 2024).

Metrics. Mathematical tasks use normalized answer accuracy and multi-hop tasks use exact match. For $k \in \{1, 2, 4\}$, pass@k is the fraction of examples with at least one correct response, while avg@k is the mean correctness across the k sampled responses. Domain scores are unweighted means over the corresponding benchmarks; Bamboogle is excluded from in-domain ablation averages. All differences are absolute percentage points.

5.2 Main Results

Tables 1–4 report the main comparisons.

9B results. At 9B, the GRPO variant gains 5.5 points on mathematics and 1.3 points on multi-hop QA over token-level GRPO; the corresponding PPO gains are 6.2 and 3.1 points. All six in-domain tasks improve under both algorithms, so the domain-level gains are not driven by a single benchmark. The largest gains are on AMC (6.1 for GRPO and 7.3 for PPO), while MuSiQue gains 2.0 under GRPO and HotpotQA gains 4.6 under PPO. Against GSPO/PPO-GSPO, the mathematics/multi-hop margins remain 3.1/2.2 points for GRPO and 3.4/3.5 points for PPO. Thus, simply aggregating the entire response does not account for the gains; adapting the aggregation span provides an additional benefit.

4B results. The 4B model exhibits the same pattern: mathematics/multi-hop gains over the token-level baselines are 4.9/1.2 points for GRPO and 6.5/2.8 points for PPO. AGREEMENT-RL also ranks first in pass@1 on every in-domain task under both algorithms. The strongest gains are 6.0 points on both AMC and NuminaMath and 2.4 points on 2WikiMultiHopQA under GRPO, and 7.2 points on AMC and 5.2 points on 2WikiMultiHopQA under PPO. Relative to GSPO/PPO-GSPO, the corresponding domain margins are 2.3/2.1 and 4.1/3.4 points. The comparable margins across scales indicate that adaptive segmentation remains useful at lower model capacity.

Computational cost. For GRPO, the average actor-update iteration takes 123.2 seconds for the 9B vanilla baseline and 143.2 seconds for AGREEMENT-RL, a 16.2% overhead. At 4B, the corresponding times are 82.3 and 99.2 seconds, a 20.5% overhead. This increment includes proxy projection, candidate scoring, dynamic programming, and joint-ratio construction.

5.3 Comparison with Proxy-PCGrad

Implementation. Exact model gradients are impractical under distributed LLM training, so we adapt PCGrad using the same Gaussian-projected logit-gradient proxy as AGREEMENT-RL. We first aggregate advantage-weighted token proxies within each domain, producing one proxy direction for mathematics and one for multi-hop QA. Let these

Model	RL	Math-only pass@1	Multi-hop-only pass@1
9B	GRPO	64.0	23.2
4B	GRPO	54.2	22.4
9B	PPO	62.2	23.2
4B	PPO	52.4	22.0

Table 5: Domain-specific training ablation (%). Math-only and multi-hop-only models are evaluated on their corresponding in-domain benchmark groups.

directions be m and q. If their inner product is nonnegative, the two domains receive equal weight. If it is negative, each domain weight is adjusted by subtracting the opposing component measured in the projected proxy space. For $d \in \{m, q\}$, the symmetric rule is

$$\begin{aligned} c &= \langle \widetilde{g}_m, \widetilde{g}_q \rangle , \\ w_d &= \begin{cases} \dfrac{1}{2}\left(1 - \dfrac{c}{\|\widetilde{g}_d\|_2^2 + \varepsilon}\right), & c < 0, \\ \frac{1}{2}, & c \geq 0, \end{cases} \\ \mathcal{L}_{\mathrm{PC}} &= w_m \mathcal{L}_m + w_q \mathcal{L}_q. \end{aligned} \tag{19}$$

This avoids storing multiple model-scale gradients. Proxy-PCGrad and AGREEMENT-RL use the same proxy, projection dimension, data, optimizer, clipping, training duration, and evaluation settings.

Results and analysis. Tables 1–4 show that Proxy-PCGrad trails vanilla RL on pass@1 in all 28 task–model–algorithm comparisons, averaging 2.9 points lower on mathematics and 2.7 points lower on in-domain multi-hop reasoning; AGREEMENT-RL exceeds it by 8.7 and 4.8 points, respectively. The two methods act at different levels. Proxy-PCGrad compresses each domain into one direction before removing opposing components, so nearly antiparallel directions can leave a weak residual and proxy error directly perturbs the weights in Equation (19). AGREEMENT-RL instead uses within-response proxy geometry to adapt temporal granularity and obtains its actor gradient from joint segment ratios. The results indicate that adaptive temporal abstraction is particularly effective in this multi-domain setting, without implying that segmentation removes every global domain conflict.

6 Analysis

6.1 Ablation Study

We isolate cross-domain interference by training the matching token-level baseline on mathematics or multi-hop QA alone. Table 5 reports performance on the corresponding in-domain benchmark group, averaging MATH L5, AMC, and NuminaMath for mathematics and 2WikiMultiHopQA, MuSiQue, and HotpotQA for multi-hop QA. This ablation is intentionally asymmetric with the main setting: each model sees only one domain, so it removes cross-domain conflict but also loses any useful transfer from the other domain.

Single-domain pass@1 gains over mixed-domain vanilla RL vary from 2.0 to 4.5 points on mathematics and from 2.2 to 4.7 points on multi-hop QA, averaging 3.4 and 3.6 points, respectively. This confirms measurable interference in naive multi-domain optimization without imposing an artificially uniform effect across settings. However, single-domain training is not a complete solution because it abandons multi-domain learning. Compared with this control, AGREEMENT-RL trained on the mixed data exceeds the math-only averages by 2.4 points on average and is 1.5 points below the multi-hop-only averages, while still improving the mixed-domain baselines in Tables 1–4. These results suggest that adaptive within-response segmentation can retain the benefits of joint training while reducing degradation associated with a fixed update granularity.

6.2 OOD Evaluation

Bamboogle is used as the OOD test because it is not included in the training mixture and was designed as a compact open-domain QA benchmark rather than another split of 2WikiMultiHopQA, MuSiQue, or HotpotQA. It therefore tests whether agreement-guided updates improve search-based generalization beyond the dataset-specific multi-hop patterns seen during training. On Bamboogle, AGREEMENT-RL improves pass@1 over vanilla RL by 0.8, 1.6, 4.8, and 4.0 points for 9B GRPO, 4B GRPO, 9B PPO, and 4B PPO, respectively. The 2.8-point average gain, together with consistent improvements over GSPO variants, indicates that agreement-guided optimization transfers beyond the training distributions.

6.3 Hyperparameter Sensitivity

Tables 6 and 7, provided in Appendix A, vary $L_{\max} \in \{256, 512, 1024\}$ and $p \in \{32, 64, 128\}$, respectively, for 4B Agreement-RL with GRPO. Across these settings, domain-averaged pass@1 varies by at most 1.4 points on mathematics and 1.5 points on multi-hop QA when changing $L_{\max}$, and by 1.3 and 2.7 points when changing p. No choice dominates every task or sampling metric, indicating limited sensitivity around the reference setting $p = 64, L_{\max} = 512$.

7 Conclusion

We introduced AGREEMENT-RL, a domain-agnostic multi-domain RL framework that discovers contiguous policy-update states through within-response token agreement and optimizes them with exact joint segment likelihood ratios. The method turns local update coherence into both a segmentation objective and an RL update unit while preserving the token-level baseline's initial gradient. Across seven evaluation sets, two model scales, and both GRPO- and PPO-style RL, AGREEMENT-RL improves pass@1 over the matching token-level baseline in all 28 comparisons. Averaged across the two algorithms, the gains are 5.9/2.2 points on 9B mathematics/multi-hop reasoning and 5.7/2.0 points at 4B, together with a 2.8-point average gain on OOD Bamboogle. Performance remains stable across the tested projection dimensions and candidate-length caps, with measured GRPO actor-update overheads of 16.2–20.5%. These results support agreement-guided temporal aggregation as a practical way to train one reasoning policy across heterogeneous domains.

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A Hyperparameter Sensitivity

We vary the maximum candidate length and projection dimension around the shared reference setting $p = 64$, $L_{\max} = 512$. The reference rows reproduce the 4B Agreement-RL (GRPO) results in Table 2.

B Projected Proxy Inner-Product Proof

For the unscaled logit proxy $q_{i,t} = e_{y_{i,t}} - \boldsymbol{\pi}_{i,t}$, the OPT closed form is

$$\|q_{i,t}\|_2^2 = 1 - 2\pi_{\theta_0}(y_{i,t} \mid x_i, y_{i,<t}) + \|\boldsymbol{\pi}_{i,t}\|_2^2, \tag{20}$$

which requires only forward-pass probabilities. For the final linear layer $z = Wh$, two logit errors u, v and hidden states h, h' satisfy

$$\langle uh^\top, vh'^\top \rangle_F = \langle u, v \rangle \langle h, h' \rangle. \tag{21}$$

The norm argument in OPT is the special case $u = v$ and $h = h'$. Equation (21) also shows why equal hidden-state norms alone do not establish cross-token angle preservation.

Let W be the policy's unembedding matrix and $h_{i,t}$ its input hidden state at token t. Using $\|\cdot\|_F$ for the Frobenius norm, the last-layer gradient is an outer product, giving

$$\|\nabla_W \log \pi_{\theta_0}(y_{i,t} \mid x_i, y_{i,<t})\|_F = \|e_{y_{i,t}} - \boldsymbol{\pi}_{i,t}\|_2 \, \|h_{i,t}\|_2. \tag{22}$$

This is the norm proportionality used by OPT. Agreement additionally requires cross-token inner products. For a candidate span, define the accumulated proxies by

$$U_i(S) = \sum_{t \in S} \xi_{i,t}, \qquad \widetilde{G}_i(S) = RU_i(S). \tag{23}$$

Let $J_{i,S}$ be a representative Jacobian mapping parameter-space changes to logit changes over span S. The local conformality condition is

$$J_{i,S} J_{i,S}^\top = c_{i,S} I, \qquad c_{i,S} > 0, \tag{24}$$

where $c_{i,S}$ is a local scale and I is the identity matrix. Given $J_{i,S}$, define each token residual exactly by

$$g_{i,t} = J_{i,S}^\top \xi_{i,t} + r_{i,t|S}, \qquad t \in S. \tag{25}$$

The second modeling assumption bounds these residuals:

$$\|r_{i,t|S}\|_2 \le \kappa \sqrt{c_{i,S}} \|\xi_{i,t}\|_2, \qquad t \in S, \qquad \kappa \ge 0. \tag{26}$$

The residual $r_{i,t|S}$ captures within-span Jacobian variation, and κ bounds its relative magnitude. Conditioned on the rollout batch, let its M nonzero token proxies be fixed independently of R. For projection distortion ϵ_{JL} and allowed failure probability η, if

$$\epsilon_{\mathrm{JL}}, \eta \in (0, 1), \qquad p \ge C \epsilon_{\mathrm{JL}}^{-2} \log(M/\eta) \tag{27}$$

for a universal constant C, then standard Gaussian concentration and a union bound give the required simultaneous event over all token pairs.

We now prove Proposition 1. Fix i, S and two positions $t < u$ in S, and abbreviate $U = \xi_{i,t}$, $V = \xi_{i,u}$, $G = g_{i,t}$, $G' = g_{i,u}$, $J = J_{i,S}$, and $c = c_{i,S}$. By Equations (24) and (25), $G = J^\top U + r$, $G' = J^\top V + r'$, and $JJ^\top = cI$. Hence

$$\begin{aligned} c^{-1}\langle G, G' \rangle = \langle U, V \rangle + c^{-1}\big(&\langle J^\top U, r' \rangle \\ &+ \langle r, J^\top V \rangle + \langle r, r' \rangle\big). \end{aligned} \tag{28}$$

Because $\|J^\top U\|_2 = \sqrt{c}\|U\|_2$ and Equation (26) holds for both tokens, Cauchy–Schwarz gives

$$\big|c^{-1}\langle G, G' \rangle - \langle U, V \rangle\big| \le (2\kappa + \kappa^2)\|U\|_2 \|V\|_2. \tag{29}$$

For a Gaussian matrix with entries $\mathcal{N}(0, 1/p)$, standard concentration implies, for any fixed U, V,

$$|\langle RU, RV \rangle - \langle U, V \rangle| \le \epsilon_{\mathrm{JL}} \|U\|_2 \|V\|_2 \tag{30}$$

with exponentially high probability in $p\epsilon_{\mathrm{JL}}^2$. A union bound over at most M^2 token pairs yields Equation (27). The triangle inequality therefore gives, simultaneously,

$$\big|\langle RU, RV \rangle - c^{-1}\langle G, G' \rangle\big| \le (\epsilon_{\mathrm{JL}} + 2\kappa + \kappa^2)\|U\|_2 \|V\|_2. \tag{31}$$

Finally, Equation (4) expresses both the ideal and projected scores as twice the sum over $t < u$. Summing Equation (31) over those pairs proves Equation (17).

C Dynamic Programming Correctness

We prove Equation (6) by induction on prefix length. The empty prefix has score zero and is optimal. Assume $F_i(q)$ is optimal for every $q < r$. Any feasible partition of $[1, r]$ ends in exactly one segment $[l, r]$. The preceding segments partition $[1, l-1]$, whose score is at most $F_i(l-1)$ by the induction hypothesis. Therefore the partition score is at most the candidate inside Equation (6). Conversely, concatenating an optimal partition for $[1, l-1]$ with $[l, r]$ constructs a feasible partition attaining that candidate. Maximizing over legal l is thus both an upper bound and achievable, proving optimality. Backpointers recover the corresponding segments.

D Joint-Ratio and Gradient Derivation

Autoregressive factorization for segment $S_{i,k} = [a, b]$ gives

$$\rho_{i,k}(\theta) = \frac{\prod_{t=a}^{b} \pi_\theta(y_{i,t} \mid x_i, y_{i,<t})}{\prod_{t=a}^{b} \pi_{\theta_0}(y_{i,t} \mid x_i, y_{i,<t})} \tag{32}$$

$$= \exp\left(\sum_{t=a}^{b} \delta_{i,t}(\theta)\right). \tag{33}$$

When clipping is inactive, differentiate $\widehat{A}_i \rho_{i,k}$ and treat the sampled advantage as constant:

$$\nabla_\theta \rho_{i,k} = \rho_{i,k} \nabla_\theta \Delta_{i,k} \tag{34}$$

$$= \rho_{i,k} \sum_{t \in S_{i,k}} \nabla_\theta \log \pi_\theta(y_{i,t} \mid x_i, y_{i,<t}), \tag{35}$$

because the old-policy term is constant. Substituting into the segment objective and evaluating at θ_0, where $\Delta_{i,k} = 0$ and

$L_{\max}$	Metric	MATH L5	AMC	Numina	2Wiki *@1 / @2 / @4*	MuSiQue	HotpotQA	Bamboogle
256	pass@k	73.8 / 83.8 / **87.4**	51.8 / 55.4 / **73.5**	39.6 / 51.6 / 56.2	27.2 / 33.0 / 35.0	7.4 / 8.6 / **11.8**	18.2 / 20.8 / 21.0	33.6 / **39.2** / 44.0
	avg@k	73.8 / 72.6 / 73.7	51.8 / 51.9 / 51.7	39.6 / 39.9 / 38.9	27.2 / 26.7 / 25.1	7.4 / 7.3 / 5.8	18.2 / 16.3 / 18.3	33.6 / 33.7 / 33.5
512	pass@k	73.0 / 82.8 / 86.2	**54.2** / 57.8 / 71.1	**41.8** / 51.0 / **58.8**	**29.2** / **35.4** / **37.4**	7.0 / 9.6 / 11.6	**21.2** / 22.0 / 22.0	**35.2** / 37.6 / **45.6**
	avg@k	73.0 / 71.7 / 72.9	**54.2** / **54.1** / **54.3**	**41.8** / **42.1** / **41.1**	**29.2** / **28.7** / **27.0**	7.0 / 6.9 / 5.4	**21.2** / **19.1** / **21.3**	**35.2** / **35.3** / **35.1**
1024	pass@k	**75.2** / **84.2** / 86.4	50.6 / **59.0** / 72.3	39.0 / **52.0** / 57.2	27.4 / 33.2 / 36.8	**7.8** / **10.0** / 11.4	19.4 / **22.6** / **23.2**	32.8 / 38.4 / 44.8
	avg@k	**75.2** / **73.9** / **75.1**	50.6 / 50.5 / 50.7	39.0 / 39.4 / 38.4	27.4 / 27.0 / 25.2	**7.8** / **7.7** / **6.1**	19.4 / 17.3 / 19.5	32.8 / 32.9 / 32.7

Table 6: Sensitivity to the maximum candidate length with $p = 64$ (%). Tasks are columns and settings are rows; each task cell lists @1 / @2 / @4. Bold marks the best value for each task, metric, and k.

p	Metric	MATH L5	AMC	Numina	2Wiki *@1 / @2 / @4*	MuSiQue	HotpotQA	Bamboogle
32	pass@k	70.0 / 82.0 / 84.4	55.4 / 56.6 / 69.9	42.8 / **53.0** / **59.4**	26.4 / 32.4 / 34.6	4.8 / 8.8 / **13.6**	18.2 / 21.0 / **24.0**	33.6 / **38.4** / 44.0
	avg@k	70.0 / 68.7 / 69.9	55.4 / 55.3 / 55.5	42.8 / 43.1 / 42.2	26.4 / 26.0 / 24.3	4.8 / 4.9 / 3.4	18.2 / 16.2 / 18.3	33.6 / 33.5 / 33.7
64	pass@k	**73.0** / **82.8** / **86.2**	54.2 / 57.8 / **71.1**	41.8 / 51.0 / 58.8	**29.2** / **35.4** / **37.4**	**7.0** / **9.6** / 11.6	**21.2** / 22.0 / 22.0	**35.2** / 37.6 / 45.6
	avg@k	**73.0** / **71.7** / **72.9**	54.2 / 54.1 / 54.3	41.8 / 42.1 / 41.1	**29.2** / **28.7** / **27.0**	**7.0** / **6.9** / **5.4**	**21.2** / **19.1** / **21.3**	**35.2** / **35.3** / **35.1**
128	pass@k	71.6 / 80.8 / 86.0	**56.6** / **59.0** / 68.7	**44.0** / 51.4 / 59.2	28.4 / 32.8 / 35.6	6.6 / 8.4 / 9.8	20.6 / **22.6** / 23.2	34.4 / 35.2 / **46.4**
	avg@k	71.6 / 70.4 / 71.5	**56.6** / **56.7** / **56.5**	**44.0** / **44.3** / **43.3**	28.4 / 27.9 / 26.3	6.6 / 6.5 / 5.0	20.6 / 18.7 / 20.5	34.4 / 34.5 / 34.3

Table 7: Sensitivity to the projection dimension with $L_{\max} = 512$ (%). Tasks are columns and settings are rows; each task cell lists @1 / @2 / @4. Bold marks the best value for each task, metric, and k.

$\rho_{i,k} = 1$, gives

$$\nabla_\theta J|_{\theta_0} = \frac{1}{B} \sum_i \frac{\widehat{A}_i}{T_i} \sum_k \sum_{t \in S_{i,k}} \nabla_\theta \log \pi_\theta(y_{i,t} \mid x_i, y_{i,<t})|_{\theta_0} \tag{36}$$

$$= \frac{1}{B} \sum_i \frac{\widehat{A}_i}{T_i} \sum_{t=1}^{T_i} \nabla_\theta \log \pi_\theta(y_{i,t} \mid x_i, y_{i,<t})|_{\theta_0} . \tag{37}$$

The last equality follows because the discovered segments partition all T_i valid response tokens exactly once. It is therefore identical to the initial gradient of the token-averaged baseline; segment ratios change only the subsequent importance weighting and clipping decisions.